

Aliphatic Polyhydrazides: A New Low Temperature Solution Polymerization

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Synopsis

High molecular weight polyhydrazides have been prepared by a new low temperature polymerization technique from equimolar amounts of a dihydrazide and a dicarbonyl chloride in dialkyl amide solvents. All-aliphatic polyhydrazides were insoluble in dimethyl sulfoxide, had polymer melt temperatures ranging from 200 to 300°C. Films could be pressed and fibers could be pulled from the melt. Aliphatic-aromatic and oxalic-aromatic polyhydrazides which undergo dehydration at different rates had polymer melt temperatures ranging from 250 to 300°C. and inherent viscosities ranging from 0.3 to 1.5. They were soluble in dimethyl sulfoxide, a solvent from which tough films could be cast.

INTRODUCTION

Carboxylic hydrazides or symmetrical carboxylic dihydrazides have been prepared by the reaction between carboxylic acids,¹ esters,² anhydrides,² or carbonyl chlorides³ with hydrazine or carboxylic hydrazides, respectively. The reaction of esters with hydrazine or with 80-100% hydrazine hydrate is generally regarded to be quantitative while the other acid derivatives frequently lead to undesirable mixtures of mono-, di-, tri-, and tetraacyl hydrazines.^{2,4-6}

This monomeric model reaction has been adapted and applied to the preparation of polymers. A French patent,⁷ as early as 1942, discloses the condensation of hydrazine or with dicarboxylic acids in alcohol and the thermal treatment of the product to yield low melting polymeric materials presumed to be low molecular weight polyhydrazides. Moldenhauer and Bock⁸ obtained in similar reaction sequences polymer which supposedly could be melt spun to fibers or melt pressed to films. Pritchard⁹ described the melt polymerization of dihydrazides in the absence of dicarboxylic acid derivatives yielding unidentified polymer. Fisher,¹⁰ and later also Korshak, Chelnokova, and Shkoline¹¹ prepared well-defined polyaminotriazoles from two moles of hydrazine and one mole of a dibasic acid at elevated temperatures and pressures. McFarlane and Miller¹³ described the preparation of a series of polyhydrazides by a high temperature solution process. Most recently, Campbell, Foldi, and Farago¹² described the

preparation of well characterized polyurelenes and of poly(acyl semi-carbazides) from stoichiometric amounts of hydrazine or dihydrazides and diisocyanates.

Since polyhydrazides may also be considered as polyamides with hydrazine, as an inorganic diamine component, it was of interest to prepare well-defined, high molecular weight aliphatic, oxalic, alicyclic, homo- and hetero-aromatic polyhydrazides and copolyhydrazides, to study their solubilities and to evaluate their utility as films and fibers. The procedure of McFarlane and Miller¹³ which involves the reaction of stoichiometric amounts of dihydrazide with dicarboxylic acids, esters, or dicarbonyl chlorides in such solvents as nitrobenzene and mixed xylenols at temperatures of 170–200°C. appeared to be suited for our purposes. In the course of this investigation, we have developed a new low temperature polymerization method involving the reaction of a dicarbonyl chloride with hydrazine or a dihydrazide at low temperatures in a dialkyl amide solvent. Thus, this report not only describes the preparation and properties of the aliphatic polyhydrazide, but in addition, compares the results of this new synthetic route with those of prior work, specifically that by McFarlane and Miller.¹³ The description of all-aromatic polyhydrazides, as a new class of polymers, has been deferred for a separate discussion.¹⁴

EXPERIMENTAL

Polymer melt temperatures (PMT) and inherent viscosities (η_{inh}) were obtained by standard methods of polymer characterization as recommended by Beaman and Cramer¹⁵ and by Sorenson and Campbell.¹⁶

The system of codes used throughout this paper simplifies the writing of structures. It is extensively used in Tables I and II. Arabic numerals indicate the number of carbon atoms in a diacid component such as dihydrazides, diacid chlorides, or diesters. The following abbreviations have been used for other diacid portions: HT = hexahydroterephthalic acid, DP = dipicolinic acid, PY = 2,6-pyrazindicarboxylic, I = isophthalic, T = terephthalic. Zero (0), as abbreviation stands for the simplest diamine, hydrazine. A polymer code such as 0-2-0-I would, therefore, represent the ordered alternating polyhydrazide derived from oxalic acid and isophthalic acid.

Intermediates, such as diacids, diacid chlorides, and dialkyl esters were obtained by conventional methods.

1. Preparation of Dihydrazides^{1-6,18}

A mixture of 1 mole of diethyl (or dimethyl) ester of a dibasic acid and 3 moles of hydrazine hydrate was refluxed in 500–1000 ml. of dry methanol for 4–24 hr. The solid dihydrazide which precipitates during the reaction is collected and recrystallized from water. In this manner, we have prepared isophthalic dihydrazide, azelaic dihydrazide and sebacic dihydrazide.

Oxalyl dihydrazide and hydrazine hydrate (Eastman Kodak), which are commercially available, were used as received.

2. Preparation of Polyhydrazides in Inert Solvents

McFarlane and Miller's method¹³ was used without change for the preparation of aliphatic polyhydrazides. A modification has also been reported by Frazer and Wallenberger.¹⁴

3. Preparation of Polyhydrazide in Amide Solvents

A solution of 0.01 mole of dicarboxylic acid dihydrazide and 20–30 ml. of *N*-methylpyrrolidone or hexamethylphosphoramide was placed in an ice-cooled reaction flask adapted with stirrer, nitrogen inlet and drying tube. To this was added portionwise over a period of ~15 min. 0.01 mole of a dicarbonyl chloride. The reaction mixture was stirred overnight, and the polymer precipitated with water. The isolated polymer was washed with ethanol and dried.

All-aliphatic polyhydrazides precipitated from the solution as the molecular weight solution viscosity increased. Aliphatic-aromatic and oxalic aromatic polyhydrazides remained in solution during the preparation.

RESULTS AND DISCUSSION

1. High Temperature Solution Preparation of Polyhydrazides

McFarlane and Miller¹³ described a high temperature solution (HTS) polymerization technique for the preparation of aliphatic and mixed aliphatic-aromatic copolyhydrazides. They reacted a diester, diacid chloride, or diacid component with hydrazine or an equimolar dihydrazide component in solvents such as mixed xyleneols or nitrobenzene at temperatures ranging from 170 to 200°C.

When these preparations¹³ were repeated, it was noted that the polymer melt temperatures of aliphatic polyhydrazides range from 185 to 328°C. (Table I) and that they were insoluble in a variety of polymer solvents including dimethyl sulfoxide. No self-supporting films and fibers were afforded by standard techniques.

The preparation of the heretofore unknown oxalic-aliphatic copolyhydrazides such as 0-2-0-6 (Table II, example 1b-1g) and the preparation of a series of aliphatic-aromatic polyhydrazides was attempted. However, only low molecular weight materials were obtained.

2. Low Temperature Solution Preparation of Polyhydrazides

Morgan, Tietz, and Sorenson,¹⁷ as well as Campbell, Foldi, and Farago¹² have studied the use of amide solvents in the preparation of condensation polymers. During this investigation, it was found that hydrazine or dihydrazides in hexamethylphosphoramide react with dicarbonyl chlorides smoothly to give high molecular weight polyhydrazides at room tempera-

TABLE I. Aliphatic Polyhydrazides

Polymer		Dicarboxylic intermediate			Method of polymerization ^b	PMT, °C.	M.P. °C.
No.	Code ^a	Acid ^a	Chloride ^a	Ester ^a	Hydrazide ^a		
1	0-2			DiPh-2	2	380	
1a	0-2-0-6			DiEt-2	6		285 ^c
b				DiEt-2	6	300	
c			6		2	295	
2a	0-2-0-9	2			9		265 ^c
b				DiEt-2	9	295	
c		2			9	320	
d				DiEt-2	9	300	
3	0-2-0-10		10		2	300	
4	0-4-0-6	4			6		285 ^c
5	0-4-0-10	10			4		300 ^c
	0-6		6		6	328	
6a	0-6-0-9	9			6		300 ^c
b			6		9	318	
7a	0-6-0-10	10			6		295 ^c
b		10			6	325	
c			10		6	308	
8	0-6-0-HT	HT			6		300 ^c
9	0-9	9			9		215 ^c
10	0-9-0-10		10		9	280	
11	0-10		10		10	280	
12	0-13				13		185 ^d
13	0-18	18			0		high ^d

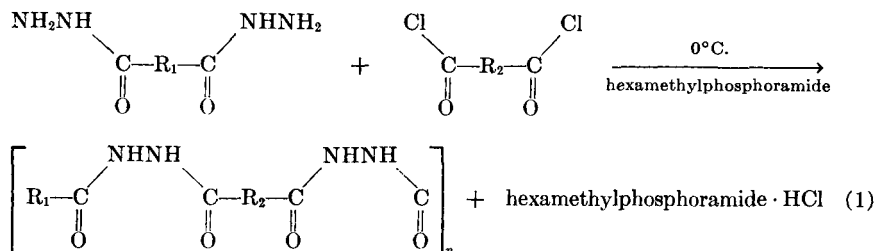
^a See experimental section for codes.^b HTS = High temperature solution polymerization; LTS = low temperature solution polymerization.^c Determination of McFarlane and Miller.¹³^d Determination of Moldehauer and Bock.^{7,8}

TABLE II
Aliphatic and Mixed Copolyhydrazides

Polymer		Dicarboxylic portion			Method of polymerization ^b		PMT, °C.	η_{inh} (DMSO)	Comments
No.	Codes ^a	Acid ^a	Chloride ^a	Ester ^a	Hydrazide ^a				
a	0-2-0-I		I		2	LTS	342	1.43	Tough, clear films cast
b		I			2	HTS	290	0.12	—
c			I		2	HTS	310	Insoluble	—
d				DiMe-I	2	HTS	250	0.04	—
e		2			1	HTS	240	Insoluble	—
f			2		1	HTS	270	0.12	—
g				DiEt-2	1	HTS	210	0.10	—
	0-2-0-DP		DP		2	LTS	350	1.01	Strong films cast
	0-2-0-PY		PY		2	LTS	380	0.33	Clear yellow films
	0-6-0-I		I		6	LTS	290	0.60	Self-supporting films
a	0-6-0-T	T			6	HTS ^c	300	—	—
b		T			6	HTS	315	Insoluble	Brittle films pressed
	0-9-0-T	T			9	HTS	320	Insoluble	Brittle films pressed
	0-10-0-I		I		10	LTS	260	1.00	Strong films cast
	0-HT-0-I		HT		1	LTS	320	0.60	Tough films cast
	0-HI-0-I		HI		1	LTS	310	0.40	Tough films cast

^a See experimental section for codes.^b See footnote b in Table I.^c Preparation of McFarlane and Miller.¹³

ture or below. *N*-Methylpyrrolidone may be used as alternate solvent to give high molecular weight polyhydrazides. This condensation, a low temperature solution (LTS) polymerization, proceeds as shown in eq. (1).



In general, optimum molecular weight and yield of polyhydrazides were obtained when the intermediates and solvents were of the highest purity and moisture vigorously excluded. Surprisingly, film-forming molecular weight polymer could also be obtained with hydrazine *hydrate* in hexamethylphosphoramide as solvent. There is no apparent explanation for this anomalous behavior.

It was, however, noted that the first members of the homologous series of aliphatic dicarbonyl chlorides gave, in decreasing order, vigorous to mild reaction with the solvent. Thus, oxalyl, malonyl, succinyl, and glutaryl chloride could not be used for the preparation of high molecular weight polyhydrazides in amide solvents. Adipyl chloride was successfully employed if it was added dropwise and over prolonged periods of time to the dihydrazide in the amide solvent solution. In order to prepare copolyhydrazides with the radicals derived from oxalyl, malonyl, succinyl, and glutaryl chloride, it was necessary to prepare the corresponding dihydrazide and to react this with a diacid chloride. Based on our experimental data it would appear that there is no other restriction on the use of dicarbonyl chlorides as well as dihydrazides.

This new synthesis, therefore, affords high-molecular weight polyhydrazides, while the McFarlane-Miller¹³ procedure does not. The copolyhydrazide, for example, consisting of alternating portions of oxalic and isophthalic acids has been prepared by 7 different approaches (Table II, 1a-1g). Only the low temperature solution method gave high molecular weight film-forming polymers. Similarly, as shown in Table II, other film-forming aliphatic or mixed polyhydrazides were obtained only by the new synthesis.

3. Solubilities and Polymer Structure

Generally all of the copolyhydrazides (aliphatic-aromatic and oxalic-aromatic) are soluble in hot, and many in cold dimethyl sulfoxide. Aliphatic polyhydrazides are less soluble. They are either swollen or partly soluble in the hot solvent. Polyhydrazides are often soluble in hexamethylphosphoramide, and occasionally in hot tetramethylene cyclic sulfone.

The polymer solubility in a series of copolyhydrazides of 0-I/0-6 ranging in mole composition from 0.00/1.00 to 1.00/0.00, was found to vary markedly as the composition was varied. Solubilities and polymer melt temperatures of these copolymers are shown in Table III. The melting point-composition curve indicates that 0-I and 0-6 are isomorphous. The 0.5/0.5 copolymer, prepared by the reaction of isophthalyl chloride with adipyl dihydrazide, is soluble in both dimethyl sulfoxide and hexamethylphosphoramide. It is by virtue of its preparation an ordered copolymer. It should be noted, however, that a random copolymer which was prepared by the reaction of equivalent amounts of isophthalyl chloride and adipyl chloride with hydrazine was insoluble in DMSO.

TABLE III
0-I/0-6 Copolymers

Composition, mole-%	η_{inh} (DMSO)	PMT °C.	Solubility		Comment
			Dimethyl sulfoxide	Hexamethyl- phosphoramide	
0.00/1.00	—	290	Insoluble	Insoluble	No films
0.33/0.67	—	293	Insoluble	Insoluble	No films
0.50/0.50	0.45	322	Soluble	Soluble	Clear films
0.67/0.33	0.49	342	Soluble	Soluble	Tough films
1.00/0.00	1.00	380	Soluble	Soluble	Tough films

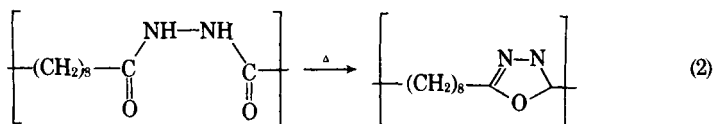
Of structural interest are the oxalic-aromatic polyhydrazides. They possess an unusually long and highly polar bishydrazide group



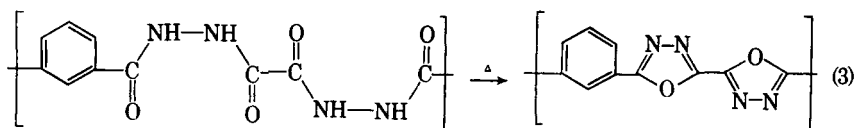
that provides the chain link within the aliphatic or aromatic polymer chain.

4. Post-Reactions and Polymer Structure

Dehydrations of a variety of monomeric polyhydrazides to give 2,5-disubstituted 1,3,4-oxadiazoles have been studied by Stolle.¹ We have applied this reaction quantitatively to prepare poly(1,3,4-oxadiazoles),¹⁴ but it seems appropriate to mention briefly the implications of this reaction which may occur as side reaction, when polyhydrazides are dried at elevated temperatures. Thus polyhydrazides may partially change into poly(1,3,4-oxadiazoles) which may be observed by slight changes in the microanalysis of dried polyhydrazides. For example, the polyhydrazide from sebacyl chloride and sebacic dihydrazide, 0-10, showed a noticeable tendency to lose water of reaction and change, at least partially into a polyoxadiazole structure [eq. (2)] upon drying above 150°C. (N₂).



Similar microanalytical evidence may be obtained for a change of oxalic acid containing polyhydrazides into the corresponding oxadiazoles. It was observed that the reaction proceeded at a faster rate. This may, at least in part, be due to the fact that a different type polyoxadiazole structure is formed during this cyclodehydration [eq. (3)] which contains adjacent



oxadiazole rings that separate the aliphatic or aromatic portion. This difference in reactivity and structure makes it advisable to distinguish aliphatic and aromatic polyhydrazides from those containing oxalic acid.

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Résumé

On a préparé des polyhydrazides de haut poids moléculaire par une nouvelle technique de polymérisation à basse température à partir de quantités équimoléculaires d'un dihydrazide et d'un chlorure de dicarbone dans des solvants dialcyl-amide. Des polyhydrazides entièrement aliphatiques sont insolubles dans le diméthylsulfoxyde, ont un point de fusion qui varie de 200° à 300°. On peut presser des films et tirer des fibres au départ de la matière fondue. Les polyhydrazides aliphatiques-aromatiques et oxaliques, qui subissent une déshydratation à vitesses différentes, ont des points de fusion qui varient de 250° à 300°C, et des viscosités inhérentes variant de 0.3 à 1.5. Ils sont solubles dans le diméthyl sulfoxyde, solvant duquel on peut obtenir des films résistants.

Zusammenfassung

Hochmolekulare Polyhydrazide wurden nach einem neuen Tieftemperatur-Polymerisationsverfahren aus äquimolaren Mengen eines Dihydrazids und eines Dicarbonylchlorids in Dialkylamid-Lösungsmitteln hergestellt. Rein aliphatische Polyhydrazide waren in Dimethylsulfoxyd unlöslich und hatten Schmelztemperaturen im Bereich von 200 bis 300°. Aus der Schmelze konnten Filme gepresst und Fasern gezogen werden. Aliphatisch-aromatische und Oxalsäure-aromatische Polyhydrazide, die mit verschiedener Geschwindigkeit dehydratisieren, besaßen Polymerschmelztemperaturen im Bereich von 250 bis 300°C und Viskositätszahlen von 0,3 bis 1,5. Sie waren in Dimethylsulfoxyd löslich und aus der Lösung konnten zähe Filme gegossen werden.

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